ARIN5204: Reinforcement Learning

1.1 Course and RL Introduction

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Some slides are from David Silver (DeepMind), Emma Brunskill (Stanford), Katerina Fragkiadaki (CMU).

Brief Self-Introduction

Long Chen (longchen@ust.hk)

- Joined HKUST CSE in 2023
- Research interests: computer vision, multimodal learning, machine learning
- Office: CYT-3003, Cheng Yu Tung Building

ARIN 5204

- First time in the 2026 spring semester
- **TAs**: Ziqi Jiang (zjiangbl@connect.ust.hk), Yanghao Wang (ywangtg@connect.ust.hk)
- Due to the Spring Festival, the lecture on Feb 16th will be moved to Feb 14th (Sat.) 10:00am, at the same classroom

For all course-related queries, please use a subject starting with [ARIN5204]

Quick Background Survey

Coding

- Python, Pytorch (or TensorFlow)

Math

- Linear algebra (inverse of a matrix, norm)
- Probability (mean/variance, sampling, Bernoulli/Gaussian distribution)
- Calculus (derivation, Taylor expansion, Lagrange multiplier)

Machine Learning and Deep Learning

- Supervised learning, loss function, gradient descent, neural network

(Deep) Reinforcement Learning?

Syllabus

RL Basics

- 1.1. Course and RL Introduction
- 1.2 Multi-armed Bandit
- 1.3 Markov Decision Processes
- 1.4 Planning by Dynamic Programming

Value-based RL

- 2.1 Model-free Prediction
- 2.2 Model-free Control
- 2.3 Value Function Approximation
- 2.4 Advanced Tricks for DQNs

Policy-based RL

- 3.1 Policy Gradient
- 3.2 Actor Critic
- 3.3 Advanced Policy Gradient
- 3.4 Continuous Control

Model-based RL

- 4.1 Imitation Learning
- 4.2 Model-based RL

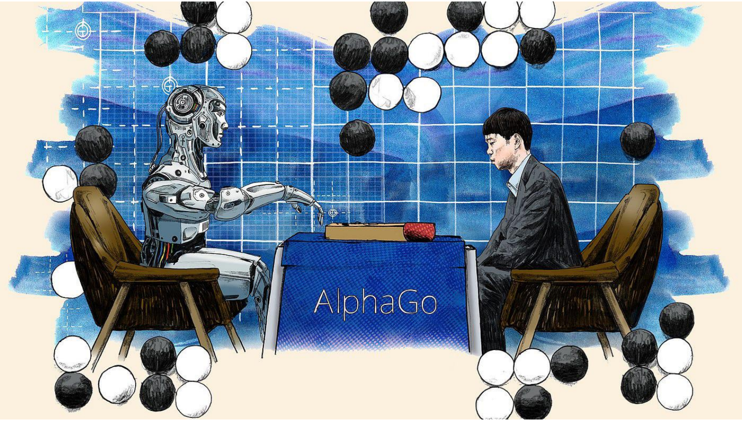
Course Information

- **Course page**: https://zjuchenlong.github.io/Teaching/26Spring_ARIN5204/
- **Slides**: Canvas (will release the slides before each class)
- **Videos**: Canvas (will release the recorded video after each class)
- **Reference books & other online resources**:
    - **books**: Reinforcement Learning: An Introduction (Sutton & Barto), ...
    - **courses**: DeepMind, UC Berkeley (CS285), Stanford (CS234), CMU (10-403), Cornell (CS 6789), Stanford (CS224R), ...

Grading Scheme

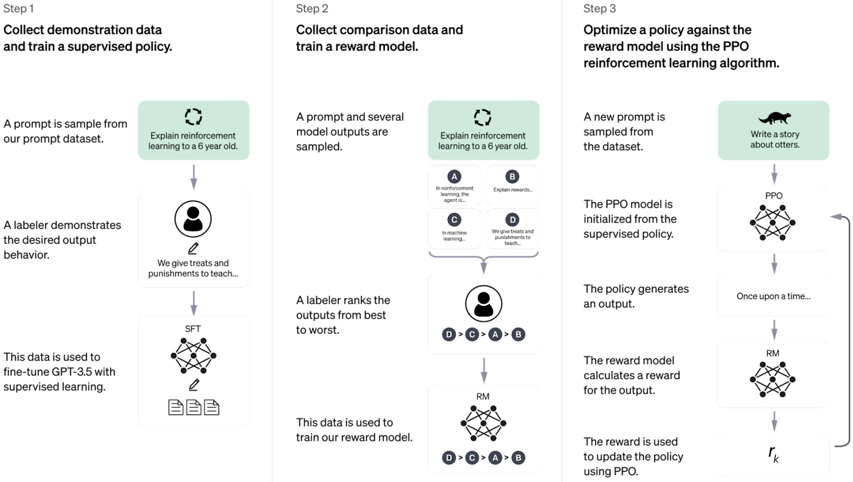
- **In-class Quiz**: 15%
  - A small question about the content in that same days lecture
  - Send your answer to TAs by email (within the same day)
- **Assignment**: 20%
  - Around 4 Python/PyTorch assignments
- **Midterm Exam**: 20%
  - Data: TBD
- **Final Exam**: 45%

Board Game Go: Beyond Human Performance

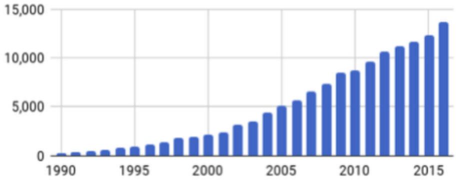


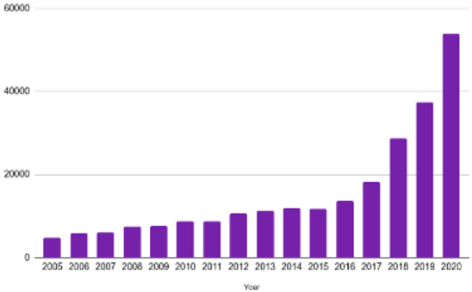
Mastering the Game of Go without Human Knowledge. Nature, 2017.

ChatGPT (https://openai.com/index/chatgpt/)



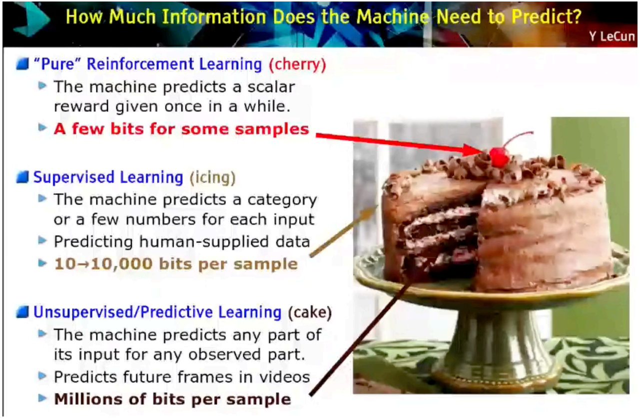
Huge Increase in Interest





Deep Reinforcement Learning that Matters. In AAAI, 2018.

But Also Contentious

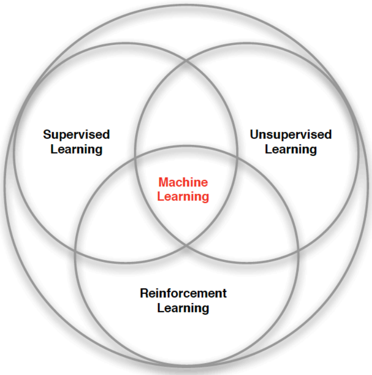


https://www.youtube.com/watch?v=Ount2Y4qxQo

Goal of the Course: Learning to Act

- Building agents that **learn** to act and accomplish **goals** in **dynamic** environments
- ... as opposed to agents that execute pre-programmed behaviors in static environments ...
- A philosophical Q: How are behaviors shaped?
  - Behavior is primarily shaped by reinforcement or free-will?
- **B.F. Skinners Statements**:
  - Behaviors that result in praise/pleasure tends to repeat.
  - Behaviors that result in punishment/pain tend to become extinct.
- https://www.youtube.com/watch?v=yhvaSEJtOV8

Machine Learning vs. Reinforcement Learning



- Supervised learning
- Unsupervised learning
- Reinforcement learning

Reinforcement Learning = Trial-and-Error Learning

Supervised Learning vs. Reinforcement Learning

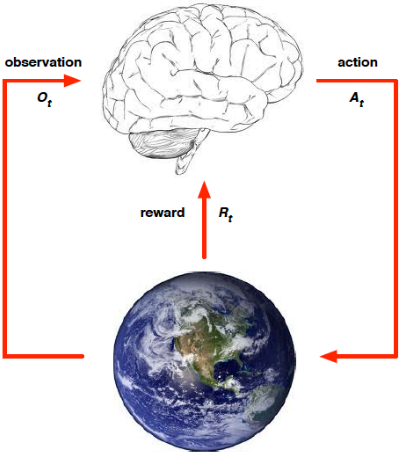
Learning from expert demonstrations (supervised learning): The expert directly suggests correct actions.

- *e.g.*, your advisor directly suggests to your ideas that are worth pursuing
- *e.g.*, for high jump, it was way easier for athletes to perfect the flop, once Fosbury showed the general trajectory

Learning from rewards (reinforcement learning). While interacting with the environment, the environment provides a signal (evaluative feedback) whether actions are good or bad.

- *e.g.*, your advisor tells you if your research ideas are worth pursuing (but does not suggest to you other ideas).
- *e.g.*, jump as high as possible. It took years to discover the Fosbury flop

Agent and Environment



- At each step t the agent:
  - Receives observation o_t
  - Receives scalar reward r_{t-1}
  - Executes action a_t
- The environment:
  - Receives action a_t
  - Emits observation o_{t+1}
  - Emits scalar reward r_t
- t increments at env. step

History and State

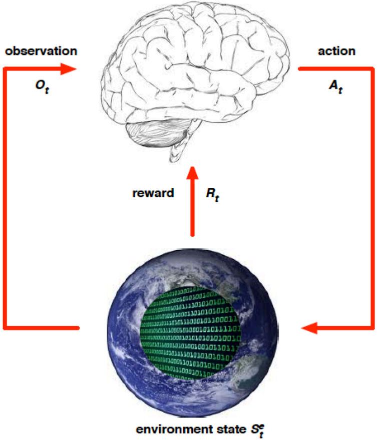
- The history is the sequence of observations, actions, rewards

$$h_t = o_1, a_1, r_1 ..., r_{t-1}, o_t, a_t, r_t$$

- *i.e.*, all observable variables up to time t
- What happens next depends on the history
  - The agent selects actions
  - The environment selects observations/rewards
- State is the information used to determine what happens next
- Formally, the state is a function of the history:

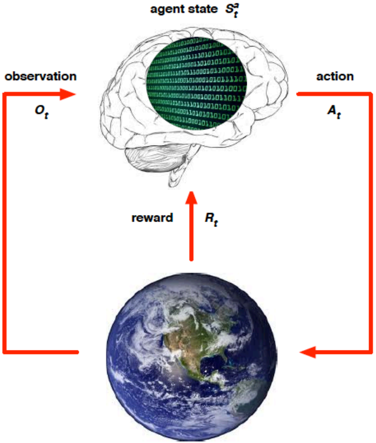
$$s_t = f(h_t)$$

Environment State



- The environment state s_t^e is the environments private representation
- *i.e.*, whatever data the environment uses to pick the next observation and reward
- The environment state is not usually visible to the agent
- Even if s_t^e is visible, it may contain irrelevant information

Agent State



- The agent state s_t^a is the agents internal representation
- *i.e.*, whatever information the agent uses to pick the next action
- *i.e.*, it is the information used by reinforcement learning algorithms
- It can be any function of history:

$$s_t^a = f(h_t)$$

Rewards

Reinforcement learning: Learning policies that maximize a reward function by interacting with the world

- A reward r_t is a scalar feedback signal
- The reward indicates how well the agent is doing at step t
- The agents job is to maximize cumulative reward

Reinforcement learning is based on the **reward hypothesis**:

- All goals can be described by the maximization of expected cumulative reward

Markov State

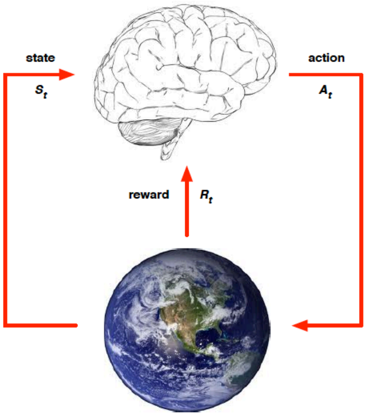
Definition

A state s_t is Markov if and only if:

$$p(s_{t+1}|s_t) = p(s_{t+1}|s_1, \dots, s_t)$$

- A Markov state contains all useful information from the history.
  - "The future is independent of the past given the present."
  - $h_{1:t} \rightarrow s_t \rightarrow h_{t+1:\infty}$. Once the state is known, the history may be thrown away
- *i.e.*, The state is a sufficient statistic of the future
- The environment state s_t^e is **Markov**, and the history h_t is **Markov**

Fully Observable Environment



- Full observability: agent directly observes environment state

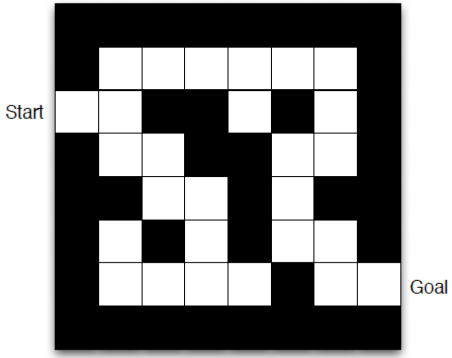
$$s_t^a = s_t^e$$

- Agent state = environment state
- Formally, it is a Markov decision process (**MDP**)

Partially Observable Environments

- Partial observability: agent indirectly observes environments:
- Now agent state $\neq$ environment state
- Formally, this is a partially observable Markov decision process (**POMDP**)
- Agent must construct its own state representation s_t^a, *e.g.*,
  - Complete history: $s_t^a = h_t$
  - Recurrent neural network: $s_t^a = \sigma(s_{t-1}^a W_s + o_t W_o)$

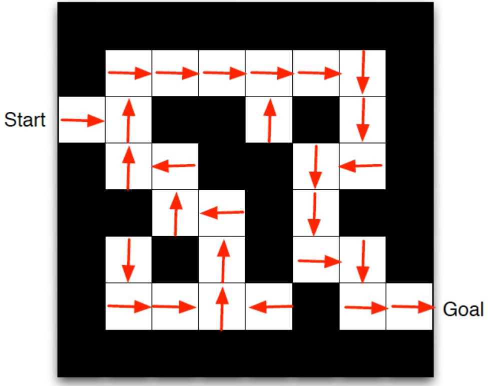
Maze Example



- **Action**: N, E, S, W
- **Reward**: -1 per time-step
- **State**: Agents location

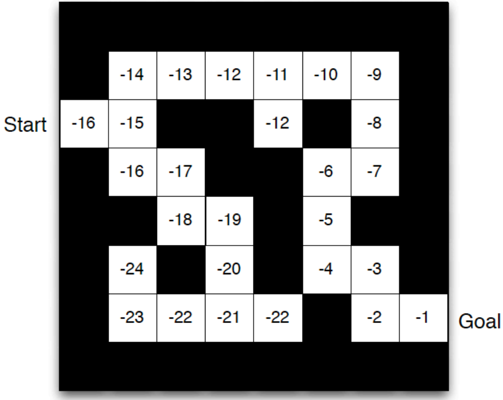
Q: full observability vs. partial observability?

Maze Example: Policy



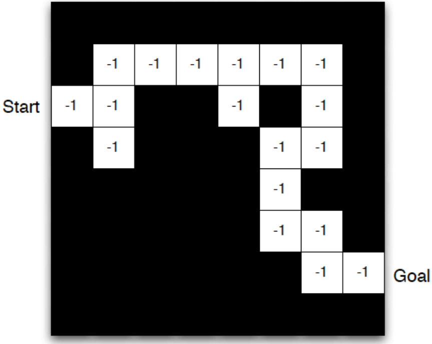
Arrows represent policy $\pi(s)$ for each state s

Maze Example: Value Function



Numbers represent value $V_\pi(s)$ of each state s

Maze Example: Model



- Agent may have an internal model of the environment
- **Dynamics**: how actions change the state
- **Rewards**: how much reward from each state
- The model may be imperfect

Major Components of an RL Agent

An RL agent may include one or more of these components:

- **Policy**: agents behavior function
- **Value function**: how good is each state and/or action
- **Model**: agent's representation of the environment

Policy

- **A policy is the agents behavior**
- It is a map from state to action, *e.g.*,
  - Deterministic policy: $a = \pi(s)$
  - Stochastic policy: $\pi(a|s) = p(a_t = a|s_t = s)$
- Q: Why deep reinforcement learning?

Value Function

- **Value function is a prediction of future reward**
- Used to evaluate the goodness/badness of states
- And therefore to select between actions, *e.g.*,

$$V_\pi(s) = \mathbb{E}_\pi\left[r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \ldots | s_t = s\right]$$

- γ is a discount factor.

Model

- **A model predicts what the environment will do next**
- $\boldsymbol{P}$ predicts the next state
- $r(\cdot)$ predicts the next (immediate) reward, *e.g.*,

$$\boldsymbol{P}(s'|s,a) = p(s_{t+1} = s'|s_t = s, a_t = a)$$

$$r(s,a) = \mathbb{E}[r_t|s_t = s, a_t = a]$$

Syllabus

Value-based

- No Policy (Implicit)
- Value Function

Policy-based

- Policy
- No Value Function

Actor-critic

- Policy
- Value Function

Model-free

- Policy and/or Value Function
- No Model

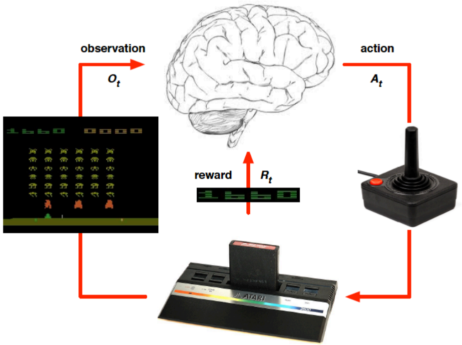
Model-based

- Policy and/or Value Function
- Model

Learning vs. Planning

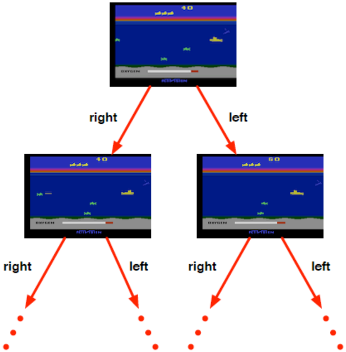
- Two fundamental problems in sequential decision making
- **Reinforcement Learning**
  - The environment is initially **unknown**
  - The agent interacts with the environment
  - The agent improves its policy
- **Planning**
  - A model of the environment is **known**
  - The agent performs computations with its model (without any external interaction)
  - The agent improves its policy

Atari Example: Reinforcement Learning



- Rules of the game are unknown
- Learn directly from interactive game-play
- Pick actions on the joystick, see pixels and scores

Atari Example: Planning



- Rules of the game are known
- Can query the emulator
  - Perfect model inside the agents brain
- If I take action a from state s:
  - What would the next state be?
  - What would the score be?
- Plan ahead to find optimal policy
  - *e.g.*, tree search

Exploration and Exploitation

- Reinforcement learning is like trial-and-error learning
- The agent should discover a good policy
- From its experiences of the environment

- Exploration finds more information about the environment
- Exploitation exploits known information to maximize reward
- It is usually important to explore as well as exploit

Characteristics of Reinforcement Learning

What makes reinforcement learning different from other machine learning paradigms?

- There is no explicit/dense supervision, only a reward signal
- Feedback is delayed, not instantaneous
- Time really matters (sequential, non i.i.d data)
- Agents actions affect the subsequent data it receives
- Active learning (collect her own data)

RL vs. Other Machine Learning

	AL Planning	Supervised Learning	Unsupervised Learning	RL	Imitation Learning
Optimization					
Learning from Experience					
Generalization					
Delayed Consequences					
Exploration					

RL vs. Other Machine Learning

	AL Planning	Supervised Learning	Unsupervised Learning	RL	Imitation Learning
Optimization	✓				
Learning from Experience					
Generalization	✓				
Delayed Consequences	✓				
Exploration					

- AI planning assumes has a model of how decisions impact environments

RL vs. Other Machine Learning

	AL Planning	Supervised Learning	Unsupervised Learning	RL	Imitation Learning
Optimization	✓				
Learning from Experience		✓			
Generalization	✓	✓			
Delayed Consequences	✓				
Exploration					

- Supervised learning has access to the correct labels

RL vs. Other Machine Learning

	AL Planning	Supervised Learning	Unsupervised Learning	RL	Imitation Learning
Optimization	✓				
Learning from Experience		✓	✓		
Generalization	✓	✓	✓		
Delayed Consequences	✓				
Exploration					

- Unsupervised learning has no labels

RL vs. Other Machine Learning

	AL Planning	Supervised Learning	Unsupervised Learning	RL	Imitation Learning
Optimization	✓			✓	
Learning from Experience		✓	✓	✓	
Generalization	✓	✓	✓	✓	
Delayed Consequences	✓			✓	
Exploration				✓	

RL vs. Other Machine Learning

	AL Planning	Supervised Learning	Unsupervised Learning	RL	Imitation Learning
Optimization	✓			✓	✓
Learning from Experience		✓	✓	✓	✓
Generalization	✓	✓	✓	✓	✓
Delayed Consequences	✓			✓	✓
Exploration				✓	

- Imitation learning typically assumes input demonstrations of good policies
- IL reduces RL to SL. **For many good reasons, IL is very popular**.

Limitations of Reinforcement Learning

- The agent should have the chance to try (and fail) enough times
- This is impossible if the episode takes too long, *e.g.*, reward = "obtain a great Ph.D."
- This is impossible when safety is a concern: we cant learn to drive via reinforcement learning in the real world, failure cannot be tolerated

Reinforcement Learning in the Real World

Q: How is the world of Alpha Go different from the real world?

- Known environment (known entities and dynamics) Vs Unknown environment (unknown entities and dynamics).
- Need for behaviors to transfer across environmental variations since the real world is very diverse
- One goal vs Many goals
- Rewards are provided automatically by an oracle environment VS rewards need to be detected
- Interactions take time: we really need intelligent exploration

Extra Reading Materials

- Chapter 1 (Introduction), Reinforcement Learning: An Introduction, Sutton & Barto.